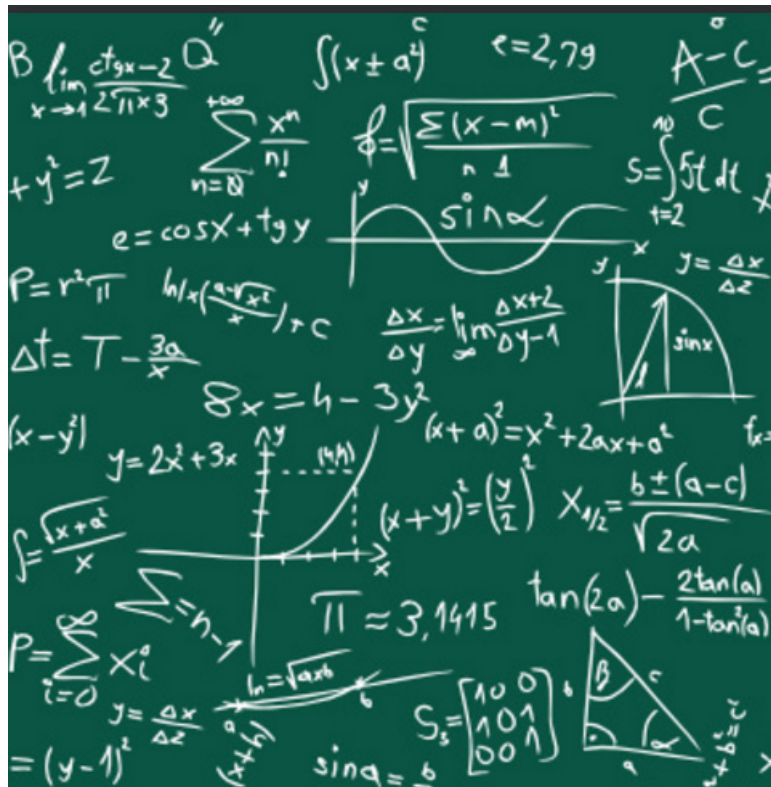


Asymptotes



- **Authors:** Fabio Furini and Gianluca Priori.
- **Book:** “Esercitazioni Analisi matematica 1”, Marco Bramanti, Esculapio.

Exercise 1

Determine the asymptotes (if any) of the following function on its natural domain

$$f(x) = \sqrt{\frac{x^3 - x}{x + 2}}$$

The function f is defined on

$$(-\infty, -2) \cup [-1, 0] \cup [1, +\infty).$$

Since

$$\lim_{x \rightarrow -2^-} f(x) = +\infty$$

the line $x = -2$ is a vertical asymptote.

For $x \rightarrow +\infty$ we have

$$\lim_{x \rightarrow +\infty} \frac{f(x)}{x} = \lim_{x \rightarrow +\infty} \sqrt{\frac{x^3 - x}{x^3 + 2x^2}} = 1$$

and

$$\lim_{x \rightarrow +\infty} f(x) - x = \lim_{x \rightarrow +\infty} \left(\sqrt{\frac{x^3 - x}{x + 2}} - x \right) \frac{\sqrt{\frac{x^3 - x}{x + 2}} + x}{\sqrt{\frac{x^3 - x}{x + 2}} + x} =$$

$$\lim_{x \rightarrow +\infty} \left(\frac{x^3 - x}{x + 2} - x^2 \right) \frac{1}{x \left(\sqrt{\frac{x^3 - x}{x^3 + 2x^2}} + 1 \right)} =$$

$$\lim_{x \rightarrow +\infty} \frac{-2x - 1}{x + 2} \frac{1}{\sqrt{\frac{x^3 - x}{x^3 + 2x^2}} + 1} = -1$$

hence we have the asymptote

$$y = x - 1, \quad x \rightarrow +\infty.$$

For $x \rightarrow -\infty$ we have

$$\lim_{x \rightarrow -\infty} \frac{f(x)}{x} = \lim_{x \rightarrow -\infty} -\sqrt{\frac{x^3 - x}{x^3 + 2x^2}} = -1$$

and

$$\lim_{x \rightarrow -\infty} f(x) + x = \lim_{x \rightarrow -\infty} \left(\sqrt{\frac{x^3 - x}{x + 2}} + x \right) \frac{\sqrt{\frac{x^3 - x}{x + 2}} - x}{\sqrt{\frac{x^3 - x}{x + 2}} - x} =$$

$$\lim_{x \rightarrow -\infty} \left(\frac{x^3 - x}{x + 2} - x^2 \right) \frac{1}{-x \left(\sqrt{\frac{x^3 - x}{x^3 + 2x^2}} + 1 \right)} =$$

$$\lim_{x \rightarrow -\infty} \frac{2x + 1}{x + 2} \frac{1}{\sqrt{\frac{x^3 - x}{x^3 + 2x^2}} + 1} = 1$$

hence we have the asymptote

$$y = -x + 1, \quad x \rightarrow -\infty.$$

Using the asymptotic expansion

$$\sqrt{1 + y} = 1 + \frac{1}{2}y + o(y), \quad y \rightarrow 0$$

and taking into account the equivalence

$$\frac{2x + 1}{x^2 + 2x} \sim \frac{2}{x}, \quad x \rightarrow \pm\infty,$$

hence

$$\frac{2x + 1}{x^2 + 2x} \sim \frac{2}{x} + o\left(\frac{1}{x}\right), \quad x \rightarrow \pm\infty,$$

we can also proceed as follows:

$$f(x) = |x| \sqrt{1 - \frac{2x + 1}{x^2 + 2x}} =$$

$$|x| \sqrt{1 - (2/x) + o(1/x)} = |x| (1 - (1/x) + o(1/x)) =$$

$$|x| - \operatorname{sgn}(x) + o(1), \quad x \rightarrow \pm\infty$$

hence we recover the oblique asymptotes $y = -x + 1$ for $x \rightarrow -\infty$ and $y = x - 1$ for $x \rightarrow +\infty$.

Exercise 2

Determine the asymptotes (if any) of the following function on its natural domain

$$f(x) = \sqrt{9x^2 + 2x + 1}$$

The function f is defined on all of $\mathbb{R}$.

For $x \rightarrow +\infty$ we have

$$\lim_{x \rightarrow +\infty} \frac{f(x)}{x} = \lim_{x \rightarrow +\infty} \sqrt{9 + 2/x + 1/x^2} = 3$$

and

$$\begin{aligned} \lim_{x \rightarrow +\infty} f(x) - 3x &= \lim_{x \rightarrow +\infty} \left(\sqrt{9x^2 + 2x + 1} - 3x \right) \frac{\sqrt{9x^2 + 2x + 1} + 3x}{\sqrt{9x^2 + 2x + 1} + 3x} = \\ \lim_{x \rightarrow +\infty} \frac{2x + 1}{3x \left(\sqrt{1 + 2/(9x) + 1/(9x^2)} + 1 \right)} &= \frac{1}{3} \end{aligned}$$

hence we have the asymptote

$$y = 3x + \frac{1}{3}, \quad x \rightarrow +\infty.$$

For $x \rightarrow -\infty$ we have

$$\lim_{x \rightarrow -\infty} \frac{f(x)}{x} = \lim_{x \rightarrow -\infty} -\sqrt{9 + 2/x + 1/x^2} = -3$$

and

$$\begin{aligned} \lim_{x \rightarrow -\infty} f(x) + 3x &= \lim_{x \rightarrow -\infty} \left(\sqrt{9x^2 + 2x + 1} + 3x \right) \frac{\sqrt{9x^2 + 2x + 1} - 3x}{\sqrt{9x^2 + 2x + 1} - 3x} = \\ \lim_{x \rightarrow -\infty} \frac{2x + 1}{-3x \left(\sqrt{1 + 2/(9x) + 1/(9x^2)} + 1 \right)} &= -\frac{1}{3} \end{aligned}$$

hence we have the asymptote

$$y = -3x - \frac{1}{3}, \quad x \rightarrow -\infty.$$

Alternatively, we can proceed as follows:

$$f(x) = 3|x|\sqrt{1 + \frac{2}{9x} + \frac{1}{9x^2}} =$$

$$3|x|\sqrt{1 + (2/9x) + o(1/x)} = 3|x|(1 + (1/9x) + o(1/x)) =$$

$$3|x| + (1/3)\text{sgn}(x) + o(1), \quad x \rightarrow \pm\infty$$

hence we recover the oblique asymptotes $y = -3x - (1/3)$ for $x \rightarrow -\infty$ and $y = 3x + (1/3)$ for $x \rightarrow +\infty$.

Exercise 3

Determine the asymptotes (if any) of the following function on its natural domain

$$f(x) = x \left(e^{-\frac{1}{x}} + \sin \frac{1}{x} \right)$$

The function f is defined for $x \neq 0$. We have

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} x e^{-\frac{1}{x}} + \lim_{x \rightarrow 0^-} x \sin \frac{1}{x} = -\infty + 0 = -\infty$$

where

$$\lim_{x \rightarrow 0^-} x e^{-\frac{1}{x}} = -\infty$$

follows from the fact that $e^{-1/x}$ is an infinite quantity of higher order than $1/x$ for $x \rightarrow 0^-$ and that $x e^{-1/x} < 0$ for $x < 0$. For the limit

$$\lim_{x \rightarrow 0^-} x \sin \frac{1}{x} = 0$$

we used the fact that the product of an infinitesimal (x) and a bounded function ($\sin(1/x)$) is an infinitesimal. The line $x = 0$ is therefore a vertical asymptote.

For $x \rightarrow \pm\infty$ we have

$$\lim_{x \rightarrow \pm\infty} \frac{f(x)}{x} = \lim_{x \rightarrow \pm\infty} e^{-\frac{1}{x}} + \sin \frac{1}{x} = 1$$

and

$$\lim_{x \rightarrow \pm\infty} f(x) - x = \lim_{x \rightarrow \pm\infty} \frac{e^{-\frac{1}{x}} - 1}{\frac{1}{x}} + \frac{\sin \frac{1}{x}}{\frac{1}{x}} =$$

$$\lim_{y \rightarrow 0^\pm} \frac{e^{-y} - 1}{y} + \lim_{y \rightarrow 0^\pm} \frac{\sin y}{y} = -1 + 1 = 0$$

hence there is the oblique asymptote

$$y = x$$

both for $x \rightarrow -\infty$ and for $x \rightarrow +\infty$.

The same result can be obtained with the expansions $e^y = 1 + y + o(y)$, $\sin y = y + o(y)$ for $y \rightarrow 0$:

$$f(x) = x(1 - 1/x + o(1/x)) + x(1/x + o(1/x)) = x - 1 + 1 + o(1) = x + o(1)$$

for $x \rightarrow \pm\infty$, which means exactly that we have the oblique asymptote

$$y = x$$

both for $x \rightarrow -\infty$ and for $x \rightarrow +\infty$.